

Tuan Tran, Ph.D. Candidate

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Education

- 2022 – current **Ph.D., University of Colorado Boulder** in Computer Science.
- 2022 – 2025 **Master., University of Colorado Boulder** in Computer Science.
GPA: **3.982/4.0**.
- 2014 – 2019 **B.Tech., Ho Chi Minh city University of Technology** in Computer Engineering.
Program in English - GPA: **8.62/10** - Rank **#1** in major, **#4** in department.
Thesis title: *Lightweight and High-throughput Anomaly-based Intrusion Detection System for Software-defined Networking (implemented on NetFPGA-10G, Xilinx)*.

Employment History

- Summer 2026 **Student Researcher**, Multimedia Streaming, ByteDance.
- 2019 – 2022 **Research Engineer**, Singapore Management University.
- Summer 2018 **Design Verification Intern**, Marvell Vietnam.

Research Publications

Conference Proceedings

2026

- 1 **Tran, Tuan**, Shirin Ebadi, S. M. H. Hosseini, Woongsub Shin, Evan Ram, Youngwook Son, Seyeon Kim, Nam Bui, Kyunghan Lee, Eric Keller, and Sangtae Ha (Aug. 2026). “DeepSFU: Scalable Deepfake Detection for Video Conferencing”. In: *Proceedings of the ACM SIGCOMM 2026 Conference. SIGCOMM '26*. Denver, CO, USA: Association for Computing Machinery, pp. 1315–1329. ISBN: 9798400724671. URL: <https://doi.org/10.1145/3789240.3829184>.
- 2 **Tran, Tuan**, S. M. H. Hosseini, Seyeon Kim, Kyunghan Lee, Nam Bui, Dirk Grunwald, and Sangtae Ha (May 2026). “eXpressSFU: Toward Super-Scalable Video Conferencing with SmartNICs”. In: *23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI' 26)*. Renton, WA: USENIX Association, pp. 2355–2369. ISBN: 978-1-939133-54-0. URL: <https://www.usenix.org/conference/nsdi26/presentation/tran>.
- 3 Ram, Evan*, **Tuan Tran***, Youngwook Son, and Sangtae Ha (2026). “RTCBench: A Scalable Benchmarking Framework for Multi-Party Video Conferencing”. In: *Proceedings of the 34th ACM International Conference on Multimedia. MM '26*. Rio de Janeiro, Brazil: Association for Computing Machinery. DOI: 10.1145/3767308.3835424. ***Both authors contributed equally to this work.**
- 4 Lee, Goodsol, Seyeon Kim, Juheon Yi, Junhong Min, **Tuan Tran**, Sangtae Ha, Kyunghan Lee, and Saewoong Bahk (May 2026). “PAVE: Mitigating Non-Congestive Delay for Seamless Video Calls over NextG Mobile Networks”. In: *IEEE INFOCOM 2026 - IEEE Conference on Computer Communications*. Tokyo, Japan: IEEE, pp. 1–10. DOI: 10.1109/INFOCOM59046.2026.11571323.

2025 and earlier

- 5 Weerakoon, Dulanga, Vigneshwaran Subbaraju, **Tuan Tran**, and Archan Misra (2022a). “SoftSkip: Empowering Multi-Modal Dynamic Pruning for Single-Stage Referring Comprehension”. In: *MM '22*. Lisboa, Portugal: Association for Computing Machinery, pp. 3608–3616. ISBN: 9781450392037. DOI: 10.1145/3503161.3548432.

- 6 Wu, Ji-Yan, Vithurson Subasharan, **Tuan Tran**, and Archan Misra (2022). "MRIM: Enabling Mixed-Resolution Imaging for Low-Power Pervasive Vision Tasks". In: *2022 IEEE International Conference on Pervasive Computing and Communications*. **PerCom '22**, pp. 44–53.  DOI: 10.1109/PerCom53586.2022.9762398.
- 7 Weerakoon, Dulanga, Vigneshwaran Subbaraju, **Tuan Tran**, and Archan Misra (2023). "Demonstrating Multi-modal Human Instruction Comprehension with AR Smart Glass". In: *2023 15th International Conference on COMMunication Systems & NETWORKS*. **COMSNETS '23**, pp. 231–233.  DOI: 10.1109/COMSNETS56262.2023.10041269.
- 8 Weerakoon, Dulanga, Vigneshwaran Subbaraju, Nipuni Karumpulli, **Tuan Tran**, Qianli Xu, U-Xuan Tan, Joo Hwee Lim, and Archan Misra (2020). "Gesture Enhanced Comprehension of Ambiguous Human-to-Robot Instructions". In: *Proceedings of the 2020 International Conference on Multimodal Interaction*. **ICMI '20**. Virtual Event, Netherlands: Association for Computing Machinery, pp. 251–259. ISBN: 9781450375818.  DOI: 10.1145/3382507.3418863.
- 9 **Tran, Minh Anh Tuan**, Trong Nhan Le, and Tan Phuong Vo (2018). "Smart-Config Wifi Technology Using ESP8266 for Low-Cost Wireless Sensor Networks". In: *2018 International Conference on Advanced Computing and Applications*. **ACOMP '18**, pp. 22–28.  DOI: 10.1109/ACOMP.2018.00012.
- 10 Tran, Ngoc Thinh, Tan Long Le, and **Minh Anh Tuan Tran** (2018). "ODL-ANTIFLOOD: A Comprehensive Solution For Securing OpenDayLight Controller". In: *2018 International Conference on Advanced Computing and Applications*. **ACOMP '18**, pp. 14–21.  DOI: 10.1109/ACOMP.2018.00011.
- 11 Ngo, Duc-Minh, Binh Tran-Thanh, Truong Dang, **Tuan Tran**, Tran Ngoc Thinh, and Cuong Pham-Quoc (2019). "High-Throughput Machine Learning Approaches for Network Attacks Detection on FPGA". In: *Context-Aware Systems and Applications, and Nature of Computation and Communication*. Ed. by Phan Cong Vinh and Abdur Rakib. Cham: Springer International Publishing, pp. 47–60. ISBN: 978-3-030-34365-1.

Journal Articles

- 1 Weerakoon, Dulanga, Vigneshwaran Subbaraju, **Tuan Tran**, and Archan Misra (2022b). "COSM2IC: Optimizing Real-Time Multi-Modal Instruction Comprehension". In: *IEEE Robotics and Automation Letters* 7.4, pp. 10697–10704.  DOI: 10.1109/LRA.2022.3194683.

Patents

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| 2026 |  Hardware-Accelerated SFU Architecture for Scalable Real-Time Media Communications. U.S. Patent Application No. 64/141,048, filed Aug 25, 2026. CU Ref. No. 2025-266-02. Status: pending. |
| |  DeepSFU: System and Method for Deepfake Detection in Video Conferencing. U.S. Patent Application No. 64/134,642, filed Aug 15, 2026. CU Ref. No. 2026-202-01. Status: pending. |
| 2025 |  SFU Architecture for Enhanced Scalability of Video Conferencing. U.S. Patent Application No. 63/869,974, filed Aug 25, 2025. CU Ref. No. 2025-266-01. Status: pending. |

Awards

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| 2025 – 2026 |  Research , Amy Barnes Frey Fellowship (\$5,000). |
| |  Research , NSDI '26 Grant (\$1,325). |
| |  Research , Conference Support Fellowship (\$1,500). |
| |  Research , Outstanding Research Paper Award (\$500). |
| |  Teaching , Outstanding Teaching Assistant Award (\$500). |

Awards (continued)

2024 – 2025	📌 Research , Society of Telecom Fellowship (\$1,000).
2023 – 2024	📌 Teaching , Outstanding Teaching Assistant Award (\$500).
	📌 Research , Departmental Summer Research Fellowship (\$6,000).
2022 – 2023	📌 Research , Faculty Advisor Fellowship (\$4,000).

Professional Activities

Reviewer	📌 IEEE Transactions on Mobile Computing .
Sub-reviewer	📌 ACM MobiCom (2025, 2026), IEEE INFOCOM (2025, 2026), ACM CoNEXT (2026).

Research Interests

Networked Systems	📌 RDMA- and DPU-accelerated infrastructure for high-throughput, low-latency streaming and AI systems.
Mobile Streaming	📌 Multipath 5G transport for low-latency, high-quality video calls at large scale.

Skills

Technical	📌 DPDK/DOCA, FPGA, RDMA, eBPF, WebRTC, Assembly.
Languages	📌 IELTS (Overall band: 7.5 ; Listening: 9.0; Reading: 7.5; Writing: 6.5; Speaking: 7.0).

References

Assoc. Prof. Sangtae Ha	📌 Department of Computer Science, University of Colorado Boulder. Director: <i>Internet Systems Lab (ISL)</i> . Email address: <i>Sangtae.Ha@Colorado.EDU</i>
Prof. Eric Keller	📌 Department of Electrical, Computer & Energy Engineering, University of Colorado Boulder. Email address: <i>Eric.Keller@Colorado.EDU</i>
Prof. Dirk Grunwald	📌 Department of Computer Science, University of Colorado Boulder. Email address: <i>Dirk.Grunwald@Colorado.EDU</i>